Mike Edison Almeida Vallejo
Independent Researcher

July 12, 2026

The Editorial Board
Journal of Cosmology and Astroparticle Physics (JCAP)

Dear Editors,

I am pleased to submit the manuscript *"Structural Self-Energy Expansion: A Minimal-Parameter Dark-Energy Framework Anchored in φ and π"* for consideration in JCAP.

The paper addresses a central question in late-time cosmology: how much of the dark-energy phenomenology can be fixed by structure rather than by free parameters. We show that the *shape* of cosmic acceleration—the equation-of-state pair (w_0, w_a), the canonical effective-field-theory coefficients, and the inflationary spectral observables—follows algebraically from two dimensionless constants, the golden ratio φ and π, with no tuned amplitude. In CMB-likelihood terms the framework is a $k = 2$ fit: it fixes four of ΛCDM's six parameters algebraically (Ω_b, Ω_c, $n_s = 1 - \varphi^{-7}$, and the derived anchor $H_0 = 3(\varphi + \pi)^2$), leaving only the primordial amplitude A_s and the optical depth τ ($k = 2$ versus $k = 6$).

The principal results that I believe merit publication are:

- The bare ratios $w_0 = -T_r/M_v = -0.840$ and $w_a = -P_{\rm sc}/K_v = -0.670$ match the DESI DR2 w_0w_aCDM posterior at the 0.24σ level (Pantheon+; 0.2–1.8σ across SN compilations), with no fitted normalization.
- A Markov-chain analysis combining DESI BAO with the ω_m-direct CMB anchor as prior yields $H_0 = 67.95 \pm 0.40\ \mathrm{km\,s^{-1}\,Mpc^{-1}}$ (0.04σ from the algebraic anchor), and the full Planck PR4 `plik` likelihood (TT, TE, EE + low-ℓ + lensing) gives $\Delta\mathrm{BIC} = -32.9$ relative to ΛCDM ($k = 2$ versus $k = 6$), with a best-fit χ^2 statistically indistinguishable from ΛCDM—the preference is driven by parsimony, not by a superior fit.
- The CMB matter density carries no fitted parameter: it follows from the forward identity $\omega_c = \mathrm{KAL}_0\, \omega_b\, n_s$, giving $\Omega_{m,\mathrm{CMB}} = \omega_m/h^2 = 0.309$ in agreement with Planck at 0.41σ and reproducing the acoustic-peak structure.
- The inflationary sector predicts $n_s = 1 - \varphi^{-7} = 0.9656$ and a tensor-to-scalar ratio $r = \varphi^{-10} = 8.1 \times 10^{-3}$, falsifiable by LiteBIRD and CMB-S4.

In keeping with the standards of the journal, the manuscript is explicit about the framework's limits. The absolute vacuum scale enters as a single measured input rather than a derived quantity; the single-sector growth amplitude S_8 remains in tension with weak-lensing surveys, relaxed only in the conditional two-sector extension; and the algebraic coefficient fixing the φ-dark-matter mass is stated as an open problem rather than a claimed result. I have deliberately separated the algebraically fixed core from these ongoing questions so that referees can assess each on its own terms.

The work is original, has not been published elsewhere, and is not under consideration by any other journal. I have no competing interests to declare. All numerical results are reproducible from the analysis scripts accompanying the manuscript.

Thank you for your time and consideration. I would be glad to respond to any questions from the editors or referees.

Sincerely,

Mike Edison Almeida Vallejo
mike.almeida1721@gmail.com